

Sandur Schist Belt, Karnataka: Mineral Prospectivity Assessment

An exploration summary of the Sandur greenstone belt from regional gravity and aeromagnetic inversion and the published geology and mining record: the known endowment, where the geophysics points beyond the existing mines, how far it can be trusted, and a phased programme to test it.

AREA 70 × 70 km around the Sandur schist belt, Ballari district, Karnataka (UTM 43N, easting 641–711 km, northing 1634–1704 km)	COMMODITIES Iron ore (principal), manganese, gold, copper
BASIS Three inversion studies (gravity, magnetics, joint; 1 km cells), measured rock samples, 15 published sources; no new data, no field visit	STATUS Desk study for screening. Nothing here is a Mineral Resource, Ore Reserve or Exploration Target in the sense of the JORC Code or NI 43-101; reserves are as reported by their sources.

Key messages

- **A proven iron district.** The belt holds about 1,876 Mt of iron ore at about 63% Fe ([GSI, Karnataka](#)); the four mines located here (NMDC's Kumaraswamy and Donimalai, SKME's Ramandurg, NEB Range) report about 370 Mt of reserves, Kumaraswamy 195 Mt at 62% Fe ([IBM 2024, Kumaraswamy](#); [GEM.wiki](#); [IBM 2018, NEB Range](#)).
- **The geophysics finds the ore's host rock.** Every iron mine lies on or within 1.4 km of a strongly magnetic column, a condition only 7%–20% of the area meets: the models map the iron formation that hosts the ore.
- **Two priority areas beyond the located mines.** R1, a remanently magnetized zone of 46 km² reaching 6 km south of Kumaraswamy, the richest part of the belt; and F1, the largest unexplained magnetic body of the belt (32 km²), in its dense centre. Further iron-formation segments (F2–F7) follow the mapped ranges.
- **Manganese, gold and copper need other methods.** Their hosts are not dense or magnetic enough to show at this scale; induced polarization, electromagnetics and geochemistry are the tools.
- **Next:** a lease and map check, then a finer inversion of the existing 37.5 m aeromagnetic grid over R1, F1 and F2, before any ground work.

≈1,876 Mt iron ore at ≈63% Fe in the belt (GSI, Karnataka)	≈370 Mt reserves of the four located mines	141 km² iron-formation horizons mapped by the magnetics; 100 km² away from the located mines
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46 km²

remanent zone R1 south of
Kumaraswamy

≈5.5 km

depth of the belt under its centre
(constrained model)

1 Commodity outlook

Table 1 · The commodities of the area and what the geophysics adds

Commodity	Known endowment	What the geophysics adds	Prospectivity	Next step
Iron ore	About 1,876 Mt at about 63% Fe in six ranges (GSI, Karnataka); four mines located here hold about 370 Mt of reserves (IBM 2024, Kumaraswamy ; GEM.wiki ; IBM 2018, NEB Range)	Maps the host iron formation: 141 km ² of horizons, 100 km ² of them more than 3 km from the located mines; a remanent zone of 46 km ² south of Kumaraswamy	High	Lease and map check of F1–F7 and R1; finer magnetic inversion
Manganese	Deogiri Formation, discontinuous over about 40 km of strike (IJERT, Sandur Mn)	Only indirectly: the iron-formation horizons mark the belts in which the manganese horizon lies	Moderate	Mapping and geochemistry along the iron-formation belts
Gold	Two low-grade occurrences: up to 0.49 g/t (Joga), 0.05–1.48 ppm (Taranagar) (Suresh & Basavanna 2014 ; Sindhuja et al. 2020)	Not resolved: dense, weakly magnetic ground beside the iron formation covers 40% of the belt	Unknown	Induced polarization and EM, soil geochemistry, structural mapping
Copper	Reconnaissance only: sporadic copper, up to 4,767 ppm in one sample, Mincheri block (NMET, Mincheri G4)	No dense or magnetic body under the block; a thin shallow dense layer only	Low	Electrical and EM methods over the block and the north-eastern belt

Prospectivity: High — the geophysics maps the host and the record proves the commodity; Moderate — the host is known but not mapped by the geophysics; Low — no supporting geophysical signal; Unknown — the host is below what these data resolve.

2 Priority areas

The areas below come from two screens of the models: iron-formation horizons (strongly magnetic columns on or beside the ridges of the belt) more than 3 km from the four located mines, and the zone where the magnetic data ask for a remanent magnetization. Priority A: a specific geophysical signal that the existing mines do not explain, beside the richest part of the belt; B: a long horizon on a mapped iron-formation range; C: small, off the belt, or a commodity the data do not resolve.

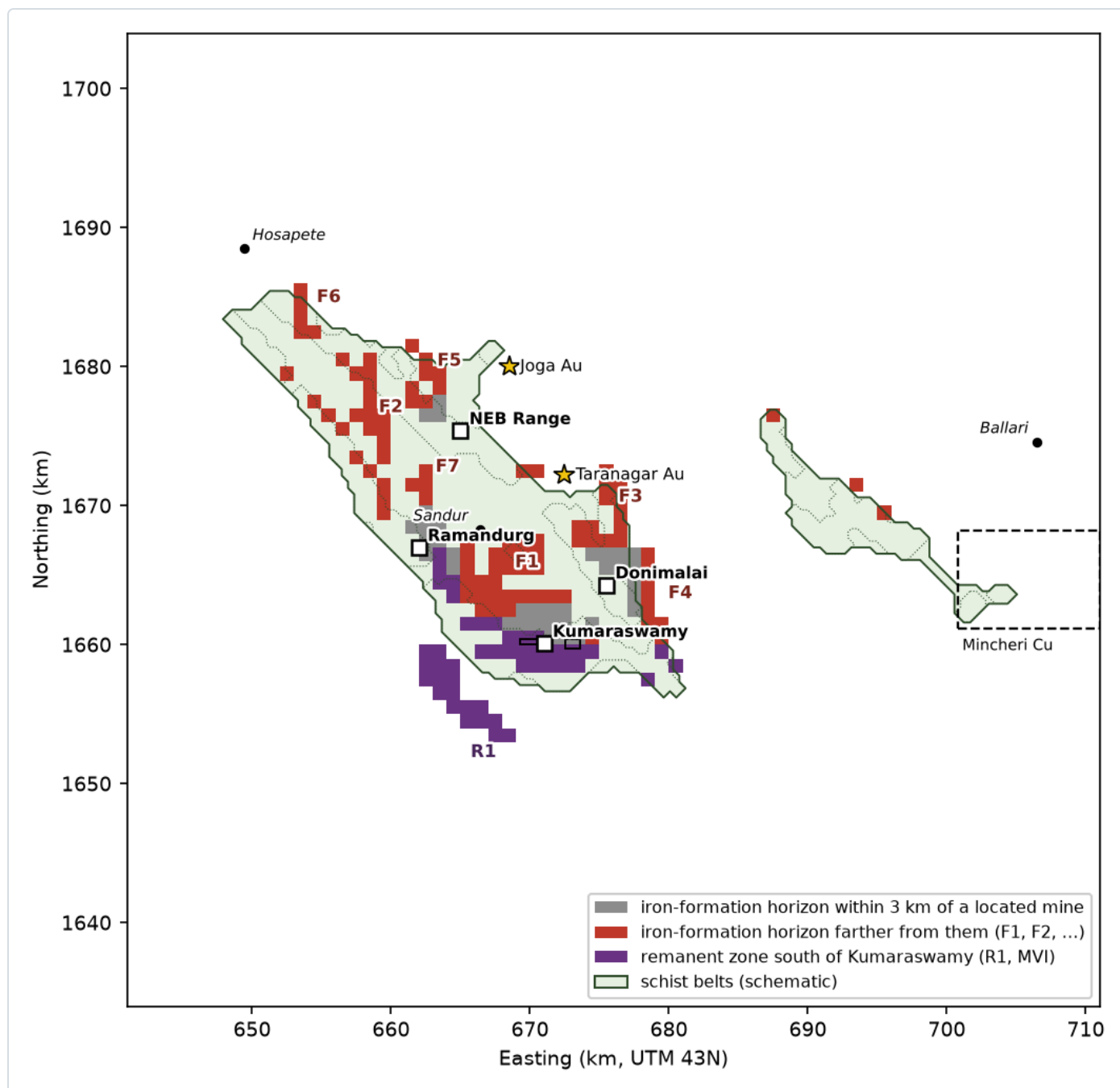


Figure 1 The priority areas: iron-formation horizons within 3 km of a located mine (grey) and farther from them (red, F1 to F7), and the remanent zone R1 south of Kumaraswamy (purple). Green: the schist belts of the schematic map (Section 3). White squares: iron-ore mines (the two small rectangles beside Kumaraswamy are its B and C development blocks); yellow stars: gold occurrences; dashed box: the Mincheri copper reconnaissance block (clipped at the edge of the area); black dots: towns.

Table 2 · Priority areas, ranked

Area	Commodity	Location (E, N km)	Size	Evidence	Confidence	Priority	Recommended action
R1	Iron	662–680, 1654–1666; at and south of Kumaraswamy	46 km ²	Magnetization not along the present field (RMS 49 → 28 nT with a vector model), 6 km beyond the Kumaraswamy B block, under the richest part of the belt	Moderate	A	Ground magnetics, oriented samples, lease check
F1	Iron	668, 1665; Sandur 4 km	32 km ²	The largest strongly magnetic body of the belt, dense (94% of its columns), in its centre under the main gravity high; folded iron formation or magnetite-bearing mafic rock	Moderate	A	Lease check; ground magnetics and mapping to tell the two apart
F2	Iron	658, 1676; western range NW of NEB Range	21 km ²	About 12 km of strongly magnetic horizon on a mapped iron-formation range (MEAI 2024)	High	B	Lease check; may be held by leases not located here

Area	Commodity	Location (E, N km)	Size	Evidence	Confidence	Priority	Recommended action
F3	Iron	675, 1669; eastern range	13 km ²	6 km of strongly magnetic horizon (mean 1.6 SI·km)	Moderate	B	Lease check
F4	Iron	679, 1662; beside the eastern range	10 km ²	9 km of strongly magnetic horizon (mean 1.2 SI·km)	Low	C	Lease check
F5	Iron	662, 1679; western range	8 km ²	5 km of strongly magnetic horizon (mean 1.5 SI·km)	Moderate	C	Lease check
F6	Iron	654, 1684; north-western end, towards Hosapete	5 km ²	4 km of strongly magnetic horizon (mean 1.0 SI·km)	Moderate	C	Lease check
F7	Iron	662, 1672; western range	4 km ²	3 km of strongly magnetic horizon (mean 0.7 SI·km)	Moderate	C	Lease check
G1	Gold	Taranagar–Joga corridor, eastern margin of the belt	—	Known occurrences in sulphidic chert and iron formation; not resolved by the potential-field models	Low	C	IP/EM, soil geochemistry, structural mapping
C1	Copper	Mincheri block and the north-eastern belt	—	Reconnaissance copper; no supporting dense or magnetic body	Low	C	IP/EM over the block

- R1 – the remanent zone.** South of Kumaraswamy the magnetic data cannot be fitted by rock magnetized along the present field; a magnetization pointing at I 74°, D –79° fits them. Hematite–magnetite ore with a remanent magnetization would explain it, and so would a magnetite-rich unit under cover. It lies beside the largest reserves of the belt and beyond the mapped ridges, which makes it the most specific iron signal of the models; it is untested.
- F1 – the central magnetic body.** The belt's centre is mapped as metavolcanic ([Mukhopadhyay & Matin 1993](#)), yet the models place a dense and strongly magnetic body there, under the main gravity high between Ramandurg and Kumaraswamy. It is either iron formation folded into the core or magnetite-bearing mafic to ultramafic rock; ground magnetics and mapping can tell which.
- F2 and the other segments** follow the two ranges of the belt. They are where iron formation is expected; their value depends on whether they are already held by the leases not located here.

3 Geological setting

The Sandur schist belt is an Archaean greenstone belt of the Dharwar craton, within the Closepet granite complex ([Maurya et al. 2023](#); [Manikyamba & Kerrich 2012](#)): a NW–SE belt about 60 km long and up to 18 km wide ([Mukhopadhyay & Matin 1993](#); [GSI, Karnataka](#)), standing as a ring of hills 900–1,050 m high, the Copper Mountain range in the east and the Sandur range in the west ([MEAI 2024](#)).

The ore horizon. The banded iron formation (Algoma type) is a chemical sediment at the top of the succession and the host of the ore. In the Sandur belt the succession comprises the Yeshwantnagar, Deogiri and Donimalai formations from the bottom up; the Deogiri Formation carries the manganese, the Donimalai Formation the iron formation ([GSI, Karnataka](#); [IJERT, Sandur Mn](#)). The western sequence runs from metabasalt at the base to manganeseiferous phyllite with chert and the iron formation at the top ([Mukhopadhyay & Matin 1993](#)); the assemblage is of the Bababudan type ([GSI, Karnataka](#)) (Table 3).

The structure. Bedding is steep and the early folds near-isoclinal, and two metasedimentary belts flank a central metavolcanic terrane ([Mukhopadhyay & Matin 1993](#)). The iron formation therefore stands in near-vertical sheets on both flanks, which is where the ranges and the mines are.

Table 3 · The succession and its ores

Unit (top to bottom)	Rocks	Ore	Source
Iron formation (Donimalai Formation, top)	Banded iron formation: amphibole, hematite, magnetite, chert; Algoma type, conformable	Iron ore by supergene enrichment; host of gold in quartz veins	(GSI, Karnataka; Turnstone, Donimalai BIF; Suresh & Basavanna 2014)
Manganiferous phyllite with chert (Deogiri Formation)	Below the iron formation	Manganese ore	(GSI, Karnataka; IJERT, Sandur Mn; Mukhopadhyay & Matin 1993)
Quartzite, carbonate, greywacke, argillite	Metasedimentary rocks of the flanks	—	(Mukhopadhyay & Matin 1993)
Metabasalt (base; the central metavolcanic terrane)	Dense, weakly magnetic; sulphidic chert interbedded	Gold in sulphidic chert	(Mukhopadhyay & Matin 1993; Sindhuja et al. 2020)
Granite and gneiss (around the belt)	The Closepet granite complex; younger granites on the eastern margin	—	(Mukhopadhyay & Matin 1993; Maurya et al. 2023)

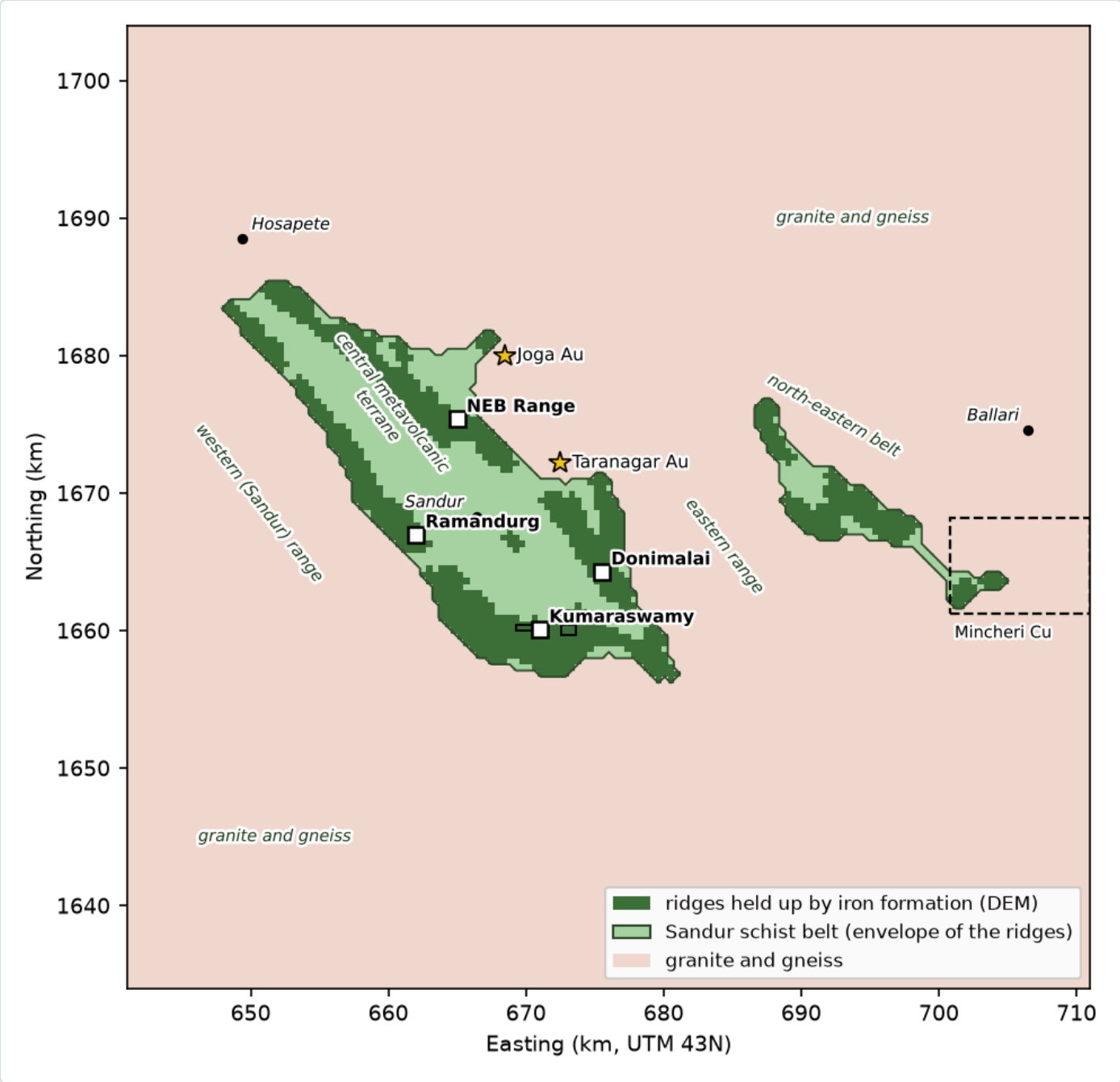


Figure 2 Schematic geology with the published localities, drawn from the 450 m DEM: ridges where the ground stands more than 80 m above the median of a 12 km window, read as the iron-formation ridges because the iron formation holds up the hills of the belt ([MEAI 2024](#)); the schist belts are the envelope of the ridges (the main belt 42 × 16 km, strike 136°); the central metavolcanic terrane after [Mukhopadhyay & Matin \(1993\)](#). A sketch for orientation, not a geological map: the mapped belt is about 60 km long, continuing north-west towards Hosapete. White squares: iron-ore mines (the two small rectangles beside Kumaraswamy are its B and C development blocks); yellow stars: gold occurrences; dashed box: the Mincheri copper reconnaissance block (clipped at the edge of the area); black dots: towns.

4 Known deposits

Table 4 · Deposits and occurrences of the area

Commodity	Locality (E, N km)	Host and form	Size, grade and depth	Source
Iron	Kumaraswamy (NMDC) 671.0, 1660.1	Supergene hematite ore in iron formation; B block 1.2 km × 650–700 m, C block 3.5 km × 450 m	Balance reserves 195.17 Mt at 62.01% Fe (1 April 2023); mean drilled depth 70 m (B), 62 m (C)	(IBM 2024, Kumaraswamy)
Iron	Donimalai (NMDC) 675.5, 1664.3	Amphibole–hematite–magnetite–chert iron formation, enriched near the surface	Reserves 127.6 Mt; capacity 7 Mt/a, 5.7 Mt mined in 2024; ore grade in the upper 170 m	(GEM.wiki; Turnstone, Donimalai BIF)

Commodity	Locality (E, N km)	Host and form	Size, grade and depth	Source
Iron	Ramandurg (SKME) 662.0, 1667.0	Iron formation	Reserves 47.7 Mt, resource 86.5 Mt; 2.6 Mt mined in 2024	(GEM.wiki)
Iron	NEB Range (ML 2150) 665.0, 1675.4	Iron formation	Balance reserve 2.6 Mt; lump and fines above 55% Fe	(IBM 2018, NEB Range)
Iron	The belt	Six ranges: Donimalai, Kumaraswamy, Ramandurg, Kanavehalli, Devagiri, Thimmappanagudi	About 1,876 Mt at about 63% Fe	(GSI, Karnataka; MEAI 2024)
Manganese	Deogiri Formation	Manganiferous phyllite with chert below the iron formation	Discontinuous over about 40 km of strike	(GSI, Karnataka; IJERT, Sandur Mn)
Gold	Joga 668.5, 1680.0	Quartz veins in the iron formation	Up to 0.49 g/t	(Suresh & Basavanna 2014)
Gold	Taranagar 672.5, 1672.2	Banded sulphidic chert	0.05 to 1.48 ppm	(Sindhuja et al. 2020)
Copper	Mincheri block (G4 reconnaissance) about 701–711, 1661–1668 (part)	Narrow ridges of banded hematite quartzite, ferruginous chert and low-grade magnetite quartzite	Sporadic copper, up to 4,767 ppm in one sample	(NMET, Mincheri G4)

Mine and occurrence positions were converted to UTM 43N (EPSG:32643). Those taken from GEM.wiki and from latitude–longitude ranges in the papers are approximate, to about 1 km, and so is the outline of the Mincheri block. The Kumaraswamy and NEB Range coordinates are as printed in the IBM inspection reports, read as UTM 43N. Reserves and production as reported by the sources, not re-estimated.

The iron ore is supergene enrichment of the iron formation near the surface: two E–W bands with a mean drilled depth of 62–70 m at Kumaraswamy ([IBM 2024, Kumaraswamy](#)), ore grade within the upper 170 m at Donimalai ([Turnstone, Donimalai BIF](#)), and hematite–magnetite ore altered from the iron formation at Obulapuram ([Bhat et al. 2025](#)). The gold occurrences are in the chemical sediments, quartz veins in the iron formation and sulphidic chert ([Suresh & Basavanna 2014; Sindhuja et al. 2020](#)); the copper of the Mincheri block is sporadic, in narrow iron-rich quartzite and chert ridges ([NMET, Mincheri G4](#)).

5 How far the geophysics can be trusted

It finds the known deposits. The test of the models is where the mines fall. Every iron mine lies on or within 1.4 km of a strongly magnetic column (≥ 0.5 SI·km integrated over depth), which only 7%–20% of the area does, and on or within 1.0 km of a dense column (≥ 0.75 g/cc·km; 11%–16% of the area). A random point would rarely do as well (Table 5).

It reproduces the belt. The dense belt of the models runs NW–SE, about 51 × 17 km; 86% of its strongly dense columns lie inside the schematic belt outline, which covers 12% of the area, and the magnetic rock reaches the surface in steep sheets at the ridges of both flanks (Figures 3 and 4), as the stratigraphy and folding predict. With the measured rock densities as bounds the belt reaches about 5.5 km under its centre, close to a published estimate of about 6 km ([Maurya et al. 2023](#)).

It does not see the ore itself. The models have 1 km cells on data continued 1 km upwards. The ore lies within 170 m of the surface, and single iron-formation layers are 100–200 m thick; a model sheet is at least one cell (1 km) wide, and the magnetic data constrain only its susceptibility × width. At Donimalai and Ramandurg the mine's own column is weakly magnetic (0.01 and 0.01 SI·km) beside a strongly magnetic one, consistent with hematite-rich enriched ore beside magnetite-bearing fresh iron formation ([Turnstone, Donimalai BIF; Bhat et al. 2025](#)), but not proven at this scale.

Table 5 · The known localities against the models

Locality	To a strongly magnetic column (km)	Area as close	To a dense column (km)	Area as close
Kumaraswamy	0.0	7%	0.0	11%
Donimalai	1.0	17%	0.0	11%
Ramandurg	1.0	17%	1.0	16%
NEB Range	1.4	20%	0.0	11%
Joga (gold)	5.0	52%	1.0	16%
Taranagar (gold)	1.0	17%	1.0	16%

Distance from each locality to the nearest 1 km column above the threshold, and the share of the area (5 km edge band left out) at least as close to such a column. Susceptibility: the magnetic study, $\beta = 1$; density: the gravity study with terrain, $\beta = 1$.

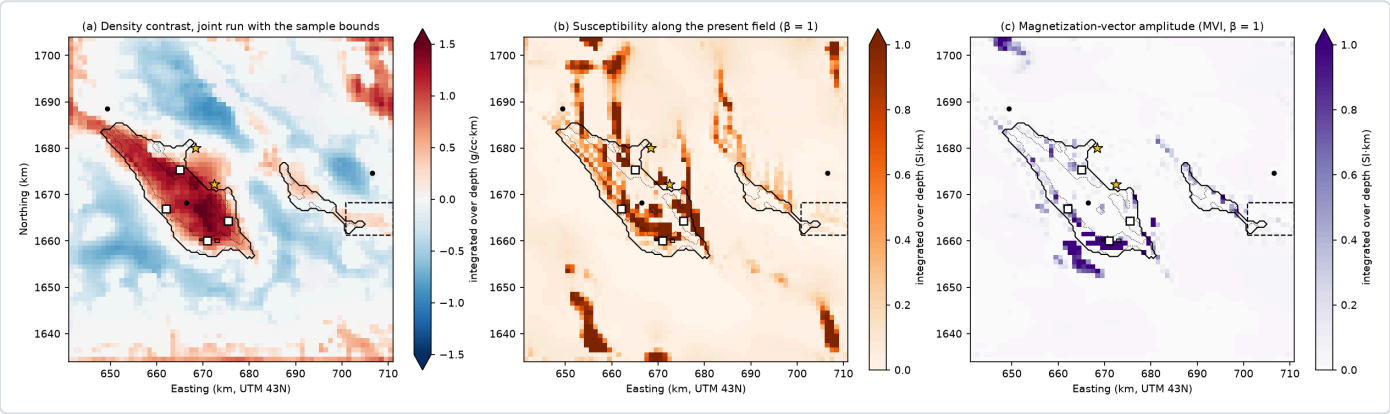


Figure 3 The three models, integrated over depth on the 1 km columns: (a) density contrast (joint inversion with the rock-sample bounds), (b) susceptibility along the present field, (c) amplitude of the magnetization vector. Black: the schematic belt outline (solid) and ridges (dotted). White squares: iron-ore mines (the two small rectangles beside Kumaraswamy are its B and C development blocks); yellow stars: gold occurrences; dashed box: the Mincheri copper reconnaissance block (clipped at the edge of the area); black dots: towns.

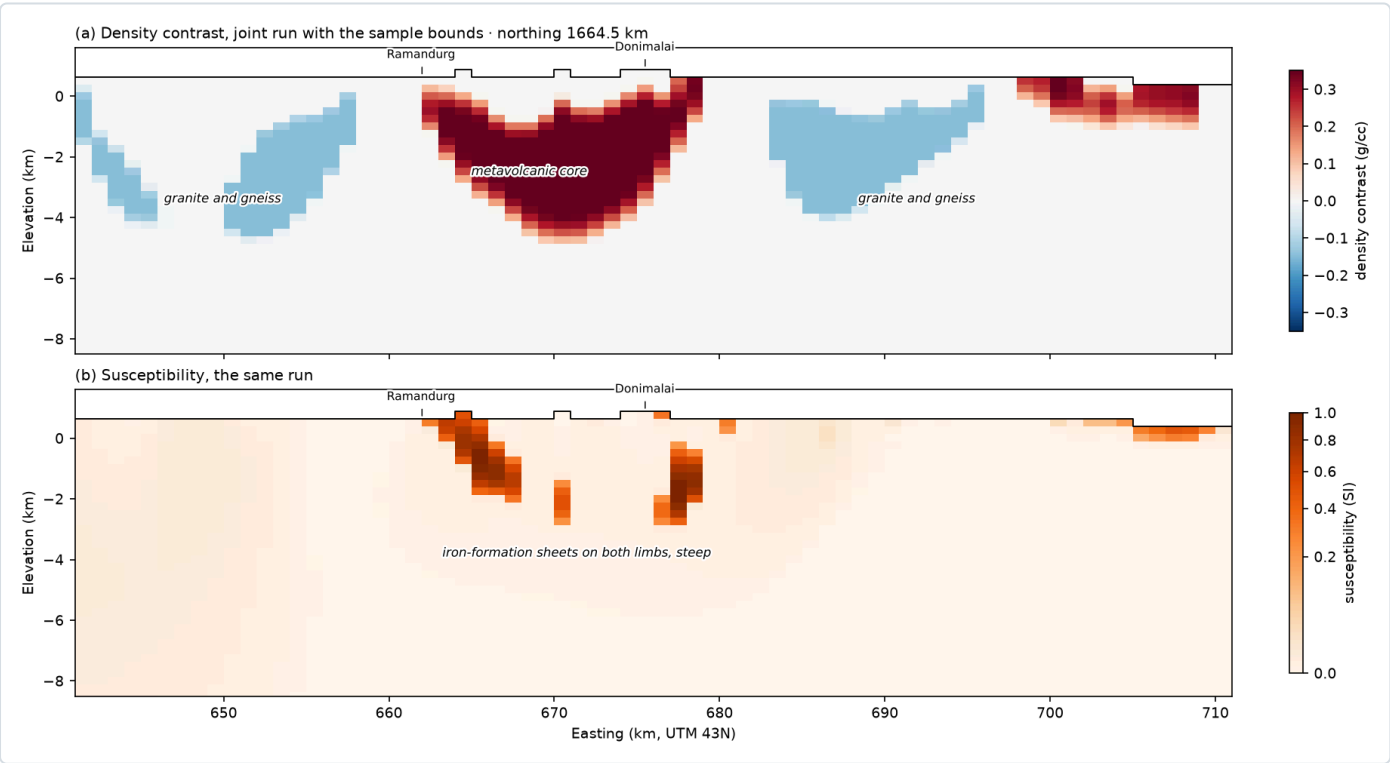


Figure 4 E–W section through the belt at northing 1664.5 km (through the main gravity high and Donimalai): density contrast and susceptibility of the joint inversion with the rock-sample bounds. A dense metavolcanic core with steep magnetic sheets on both flanks; the black line is the ground.

6 Manganese, gold and copper

- **Manganese** lies in the Deogiri Formation below the iron formation, over about 40 km of strike ([GSI, Karnataka; IJERT, Sandur Mn](#)). Manganese oxide in phyllite is neither dense nor magnetic enough at 1 km cells; the iron-formation horizons of Section 2 mark the belts in which it lies.
- **Gold** is in sulphidic chert and iron formation ([Suresh & Basavanna 2014; Sindhuja et al. 2020](#)), magnetite-poor rock that is dense but weakly magnetic. Such ground beside the iron formation covers 40% of the belt and includes Taranagar but not Joga: the screen does not single out targets. Induced polarization, EM and soil geochemistry along the eastern margin between Taranagar and Joga are the next step.
- **Copper**. Under the Mincheri block the models hold only a thin, shallow dense layer (up to 0.36 g/cc·km, centred 0.7 km deep) and weak magnetization (up to 0.20 SI·km), consistent with the narrow low-grade ridges mapped there ([NMET, Mincheri G4](#)) and with no large body. Copper sulphides need electrical and EM methods.

7 Recommended programme

Table 6 · A phased programme with a decision after each phase

Phase	Work	Decision it informs
1 · Desk (weeks)	Check F1–F7 and R1 against the 1:50,000 geology (GSI Bhukosh) and the lease records; invert the 37.5 m magnetic grid without upward continuation over R1, F1 and F2 (cloud compute, a few hours)	Which areas are open and still anomalous at 100 m cells
2 · Ground (months)	Ground magnetics and geological mapping over R1, F1 and F2; oriented samples for remanence; rock-property samples	Whether R1 and F1 are ore-bearing iron formation
3 · Follow-up	Induced polarization / resistivity and EM over the Taranagar–Joga corridor and the Mincheri block with soil geochemistry; drilling of the best iron area	Drill targets; gold and copper go/no-go

The first phase uses data already in hand. The aeromagnetic survey was flown 80 m above the ground on lines 300 m apart, oriented NE–SW across the strike of the belt, and gridded at 37.5 m: inverted without upward continuation on a finer mesh, it can resolve the iron-formation layers that the regional 1 km models merge, and test R1 and F1 at a cost of hours of cloud computing.

8 Risks and limitations

Table 7 · Risks and their mitigation

Risk	Effect	Mitigation
Lease status unknown	F1–F7 may lie in existing leases (the belt has six iron ranges, four located here)	Check against the IBM lease records before any spend
Resolution	1 km cells on data continued 1 km upwards: the ore (within 170 m of the surface) and single iron-formation layers (100–200 m) are not resolved	Finer inversion of the 37.5 m survey grid
Remanence untested	R1 rests on a magnetization direction no palaeomagnetic data confirm	Oriented samples at Kumaraswamy and over R1
Depth	The depth of the belt follows the inversion's assumptions	Use the depths as ranges, not values
Gold and copper	Sulphide hosts are not seen by gravity and magnetics	Electrical and EM methods
Data	Four mines and two gold occurrences located, to about 1 km; reserves as reported by the sources	Full drilling and lease data from the operators or IBM

Appendix: method notes

- **Models**. Density from the joint gravity–magnetic inversion with the measured rock densities as bounds (–0.15 to +0.35 g/cc around 2.66 g/cc) and a depth weighting of $\beta = 0.5$; susceptibility and the magnetization-vector

amplitude from the magnetic study ($\beta = 1$); 1 km $\times$ 1 km $\times$ 250 m cells. Details: the gravity, magnetic and joint inversion reports.

- **Iron-formation horizons.** 1 km columns with integrated susceptibility ≥ 0.5 SI·km within 1.5 km of a schematic ridge or inside a schist belt; F-segments: those more than 3 km from the four located mines, at least 3 connected columns.
- **R1.** Columns with integrated MVI amplitude ≥ 0.5 SI·km within 2 km of a station the induced model underfits by more than 150 nT, within 10 km of Kumaraswamy.
- **Gold screen.** Within 2 km of a dense column (≥ 0.75 g/cc·km), within 3 km of an iron-formation horizon, integrated susceptibility < 0.2 SI·km.
- **Schematic map.** Drawn from the 450 m DEM of the inputs; no published map is copied (`karnataka_inputs/shared/literature.py`).
- **Scripts.** `examples/output/karnataka_minerals/scripts/make_figures.py` (screens), `validation.py` (the localities against the models), `build_report.py`.

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Mine and occurrence positions were converted to UTM 43N (EPSG:32643). Those taken from GEM.wiki and from latitude–longitude ranges in the papers are approximate, to about 1 km, and so is the outline of the Mincheri block. The Kumaraswamy and NEB Range coordinates are as printed in the IBM inspection reports, read as UTM 43N.

This assessment screens publicly available data and published information; it has not been verified in the field. It is not a Mineral Resource or Ore Reserve estimate, nor an Exploration Target under the JORC Code or NI 43-101, and should not be used as one. Reserve and production figures are those reported by the cited sources.

GeoInv3D · Generated by build_report.py; the model values come from the runs of the three inversion reports, the published values from the sources cited.